

Rezolvare — Bilet de examen SAE, sesiunea toamnă 2026

1. $G(s) = \frac{9}{s(s+3)}, T_e = 1s$

a) $H(z)$ – metoda trapezului (Tustin)

$$s = \frac{2}{T_e} \cdot \frac{z-1}{z+1} = 2 \frac{z-1}{z+1} \quad s+3 = \frac{2(z-1)+3(z+1)}{z+1} = \frac{5z+1}{z+1}$$

$$G(z) = \frac{9}{s(s+3)} = \frac{9(z+1)^2}{2(z-1)(5z+1)} = \frac{9z^2+18z+9}{10z^2-8z-2}$$

$$H(z) = \frac{G(z)}{1+G(z)} = \frac{9z^2+18z+9}{(10z^2-8z-2)+(9z^2+18z+9)}$$

$$H(z) = \frac{9z^2+18z+9}{19z^2+10z+7}$$

b) $G(z)$ – ecuația cu diferențe

$$G(s) = \frac{9}{s(s+3)} \Rightarrow \ddot{y} + 3\dot{y} = 9u$$

$$\dot{y} \approx \frac{y[k]-y[k-1]}{T_e} = y[k]-y[k-1] \quad \ddot{y} \approx \frac{y[k]-2y[k-1]+y[k-2]}{T_e^2} = y[k]-2y[k-1]+y[k-2]$$

$$(y[k]-2y[k-1]+y[k-2])+3(y[k]-y[k-1])=9u[k]$$

$$4y[k]-5y[k-1]+y[k-2]=9u[k]$$

$$q^{-1}y[k]=y[k-1]: \quad (4-5q^{-1}+q^{-2})y[k]=9u[k] \Rightarrow G(q) = \frac{9}{4-5q^{-1}+q^{-2}} \xrightarrow{q \rightarrow z}$$

$$G(z) = \frac{9z^2}{4z^2-5z+1} = \frac{9z^2}{(z-1)(4z-1)}$$

c) Eroare staționară

$$G(z) = \frac{9(z+1)^2}{2(z-1)(5z+1)}$$

$$K_p = \lim_{z \rightarrow 1} G(z) = \infty$$

$$K_v = \frac{1}{T_e} \lim_{z \rightarrow 1} (z-1)G(z) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{9(z+1)^2}{2(z-1)(5z+1)} = \lim_{z \rightarrow 1} \frac{9(z+1)^2}{2(5z+1)} = \frac{9 \cdot 4}{2 \cdot 6} = \frac{36}{12} = 3$$

$$K_a = \frac{1}{T_e^2} \lim_{z \rightarrow 1} (z-1)^2 G(z) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{9(z+1)^2}{2(5z+1)} = 0$$

$$e_{ss}(\text{treaptă}) = \frac{1}{1+K_p} = 0 \quad e_{ss}(\text{rampă}) = \frac{1}{K_v} = \frac{1}{3} \approx 0.33 \quad e_{ss}(\text{parabolă}) = \frac{1}{K_a} = \infty$$

d) Răspuns la treaptă unitară ($\mathcal{Z}^{-1}$)

$$H(z) = \frac{Y(z)}{R(z)} = \frac{9z^2+18z+9}{19z^2+10z+7} \Rightarrow Y(z) = H(z) \cdot R(z)$$

$$R(z) = \frac{z}{z-1}$$

$$Y(z) = \underbrace{\frac{9z^2+18z+9}{19z^2+10z+7}}_{H(z)} \cdot \underbrace{\frac{z}{z-1}}_{R(z)} = \frac{(9z^2+18z+9) \cdot z}{(19z^2+10z+7) \cdot (z-1)}$$

$$Y(z) = \frac{z(9z^2+18z+9)}{(z-1)(19z^2+10z+7)}$$

$$z_{1,2} = \frac{-10 \pm \sqrt{100 - 4 \cdot 19 \cdot 7}}{38} = -0.26 \pm j0.55 = \alpha \pm j\beta = r e^{\pm j\theta}$$

$$r = |z_{1,2}| = \sqrt{\alpha^2 + \beta^2} = \sqrt{7/19} = 0.61$$

$$\theta = \arg z_{1,2} = 180^\circ - \arctan \frac{\beta}{|\alpha|}$$

$$\theta = 180^\circ - \arctan \frac{0.55}{0.26} = 180^\circ - \arctan(2.08) = 180^\circ - 64.31^\circ$$

$$\theta = 115.69^\circ$$

$$\frac{Y(z)}{z} = \frac{9z^2 + 18z + 9}{(z-1)(19z^2 + 10z + 7)} = \frac{A}{z-1} + \frac{Bz + C}{19z^2 + 10z + 7}$$

$$9z^2 + 18z + 9 = A(19z^2 + 10z + 7) + (Bz + C)(z-1)$$

$$z = 1 : 36 = 36A \Rightarrow A = 1 \quad \text{coef. } z^2 : 19A + B = 9 \Rightarrow B = -10 \quad \text{termen liber : } 7A - C = 9 \Rightarrow C = -2$$

$$19z^2 + 10z + 7 = 19(z - z_1)(z - z_2) = 19[z - (\alpha + j\beta)][z - (\alpha - j\beta)] \\ = 19[(z - \alpha)^2 - (j\beta)^2] = 19[(z - \alpha)^2 + \beta^2] = 19[z^2 - 2\alpha z + \underbrace{\alpha^2 + \beta^2}_{r^2}]$$

$$19z^2 + 10z + 7 = 19(z^2 - 2r \cos \theta + r^2), \quad r \cos \theta = \alpha = -0.26, \quad r \sin \theta = \beta = 0.55$$

$$\mathcal{Z}\{r^k \cos k\theta\} = \frac{z(z - r \cos \theta)}{z^2 - 2rz \cos \theta + r^2} \quad \mathcal{Z}\{r^k \sin k\theta\} = \frac{zr \sin \theta}{z^2 - 2rz \cos \theta + r^2}$$

$$Y(z) = \frac{z}{z-1} + \frac{z(Bz + C)}{19(z^2 - 2rz \cos \theta + r^2)}$$

$$Bz + C = B(z - r \cos \theta) + \underbrace{[Br \cos \theta + C]}_D$$

$$D = Br \cos \theta + C = (-10)(-0.26) + (-2) = 2.63 - 2 = 0.63$$

$$\frac{z(Bz + C)}{19(z^2 - 2rz \cos \theta + r^2)} = \frac{B}{19} \cdot \frac{z(z - r \cos \theta)}{z^2 - 2rz \cos \theta + r^2} + \frac{D}{19} \cdot \frac{z}{z^2 - 2rz \cos \theta + r^2}$$

$$\frac{B}{19} = \frac{-10}{19} = -0.53$$

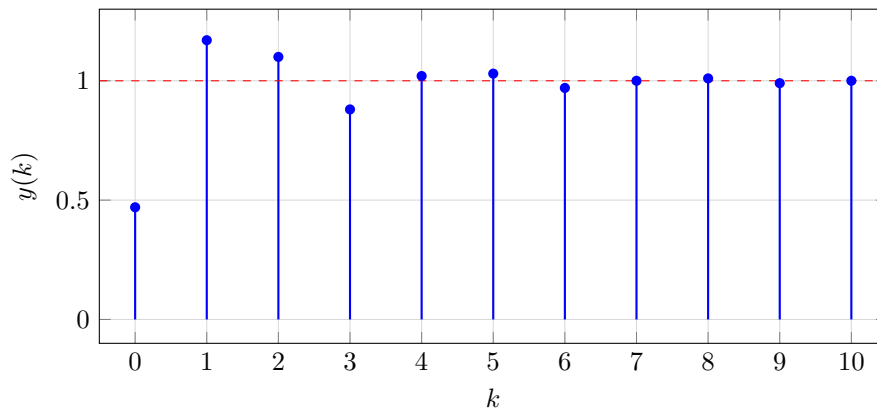
$$\frac{D}{19} z = \frac{D}{19 r \sin \theta} \cdot (z r \sin \theta)$$

$$\frac{D}{19 r \sin \theta} = \frac{0.63}{19 \cdot 0.55} = \frac{0.63}{10.39} = 0.06$$

$$Y(z) = \frac{z}{z-1} \underbrace{-0.53}_{B/19} \frac{z(z - r \cos \theta)}{z^2 - 2rz \cos \theta + r^2} + \underbrace{0.06}_{D/(19 r \sin \theta)} \frac{z r \sin \theta}{z^2 - 2rz \cos \theta + r^2}$$

$$y(k) = 1 - 0.53 r^k \cos(k\theta) + 0.06 r^k \sin(k\theta), \quad k \geq 0$$

k	0	1	2	3	4	5	6	7	8	9	10
$y(k)$	0.47	1.17	1.10	0.88	1.02	1.03	0.97	1.00	1.01	0.99	1.00



e) Caracteristici de frecvență (transformarea w)

$$z = \frac{1 + \frac{T_e}{2}w}{1 - \frac{T_e}{2}w} \quad T_e=1 \quad \frac{1 + 0.5w}{1 - 0.5w} = \frac{N}{D}, \quad N = 1 + 0.5w, \quad D = 1 - 0.5w \Rightarrow z + 1 = \frac{2}{D}$$

$$H(z) = \frac{9(z+1)^2}{19z^2 + 10z + 7} \quad 9(z+1)^2 = \frac{36}{D^2} \quad 19z^2 + 10z + 7 = \frac{19N^2 + 10ND + 7D^2}{D^2} = \frac{4w^2 + 12w + 36}{D^2}$$

$$H(w) = \frac{36}{4w^2 + 12w + 36} = \frac{9}{w^2 + 3w + 9}$$

$$H(w) = \frac{9}{w^2 + 3w + 9} = \frac{1}{1 + \frac{1}{3}w + \frac{1}{9}w^2} \stackrel{!}{=} \frac{K}{1 + 2\alpha Tw + T^2 w^2}$$

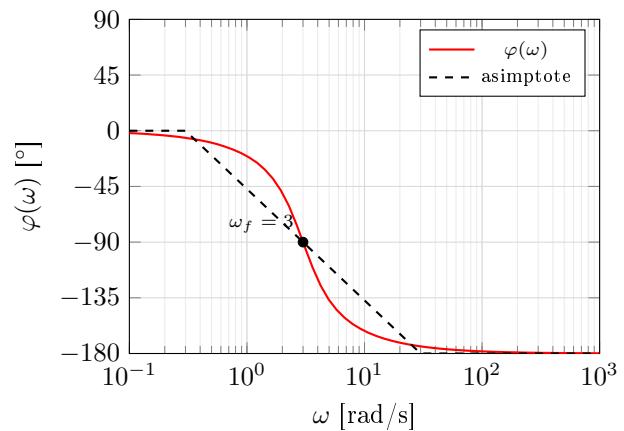
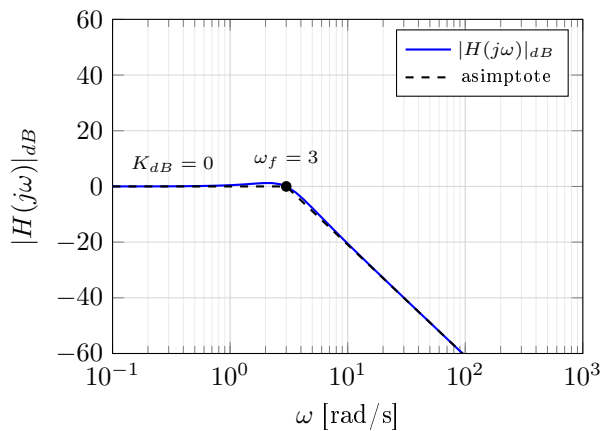
$$\text{termen liber: } 1 = K \Rightarrow K = 1$$

$$\text{coef. } w^2: T^2 = \frac{1}{9} \Rightarrow T = \frac{1}{3}$$

$$\text{coef. } w: 2\alpha T = \frac{1}{3} \Rightarrow \alpha = \frac{1}{2T} \cdot \frac{1}{3} = \frac{1}{2 \cdot \frac{1}{3}} \cdot \frac{1}{3} = \frac{1}{2}$$

$$T = \frac{1}{3} = 0.33 \text{ s} \quad \alpha = 0.5 \quad K = 1 \Rightarrow K_{dB} = 0 \text{ dB} \quad \omega_f = \frac{1}{T} = 3 \text{ rad/s}$$

- amplitudine: $K_{dB} = 0 \text{ dB}$ pentru $\omega < \omega_f$; pantă $-40\text{dB}/\text{decadă}$ pentru $\omega > \omega_f$; frângere $(\omega_f, K_{dB}) = (3, 0)$.
- fază: 0° pentru $\omega < \omega_f/10 = 0.3$; dreaptă (în $\lg \omega$) până la -180° la $10\omega_f = 30$; -90° la $\omega = \omega_f = 3$.



f) Model cu variabile de stare

$$H(z) = \frac{0.47z^2 + 0.95z + 0.47}{z^2 + 0.53z + 0.37} = D + \frac{b'_1 z + b'_0}{z^2 + a_1 z + a_0}$$

$$D = 0.47 \quad b'_1 = 0.95 - D \cdot 0.53 = 0.70 \quad b'_0 = 0.47 - D \cdot 0.37 = 0.30$$

$$x_1(k+1) = x_2(k) \quad x_2(k+1) = -0.37x_1(k) - 0.53x_2(k) + r(k)$$

$$y(k) = 0.30x_1(k) + 0.70x_2(k) + 0.47r(k)$$

$$\Phi = \begin{pmatrix} 0 & 1 \\ -0.37 & -0.53 \end{pmatrix}, \quad \Gamma = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C = (0.30 \quad 0.70), \quad D = 0.47$$

$$C(zI - \Phi)^{-1}\Gamma + D = \frac{0.47z^2 + 0.95z + 0.47}{z^2 + 0.53z + 0.37} = H(z) \checkmark$$

$$2. \quad G_d(z) = \frac{0.37k(z + 0.72)}{z^2 - 1.37z + 0.37}, \quad T_e = 1\text{s}$$

$$z^2 - 1.37z + 0.37 = (z - 1)(z - 0.37)$$

Poli: $p_1 = 1, p_2 = 0.37$. Zero: $z_0 = -0.72$.

a) Poli, zerouri, ramuri

$n = 2, m = 1 \Rightarrow \text{nr. ramuri} = 2$.

Ramuri din $p_1 = 1, p_2 = 0.37$; se termină în $z_0 = -0.72$ și în $z \rightarrow \infty$.

b) Axa reală

$$(p_2, p_1) = (0.37, 1) \quad (-\infty, z_0) = (-\infty, -0.72)$$

c) Asimptote

$$n - m = 1, \quad \theta = \frac{180^\circ(2i + 1)}{n - m} \Big|_{i=0} = 180^\circ \quad \sigma_a = \frac{\sum p_i - \sum z_i}{n - m} = \frac{1.37 + 0.72}{1} = 2.09$$

1 asimptotă, 180° , $\sigma_a = 2.09$.

d) Puncte de ramificație

$$1 + G_d(z) = 0 \Rightarrow k(z) = -\frac{N(z)}{D(z)}$$

$$N(z) = (z - 1)(z - 0.37) = z^2 - 1.37z + 0.37 \quad D(z) = 0.37(z + 0.72) = 0.37z + 0.26$$

$$N'(z) = 2z - 1.37 \quad D'(z) = 0.37$$

$$\frac{dk}{dz} = -\frac{N'(z)D(z) - N(z)D'(z)}{D(z)^2} = 0 \Leftrightarrow N'(z)D(z) - N(z)D'(z) = 0$$

$$N'(z)D(z) = (2z - 1.37)(0.37z + 0.26)$$

$$= 2z \cdot 0.37z + 2z \cdot 0.26 - 1.37 \cdot 0.37z - 1.37 \cdot 0.26$$

$$= 0.74z^2 + 0.53z - 0.50z - 0.36 = 0.74z^2 + 0.02z - 0.36$$

$$N(z)D'(z) = 0.37 \cdot (z^2 - 1.37z + 0.37) = 0.37z^2 - 0.50z + 0.14$$

$$N'(z)D(z) - N(z)D'(z) = (0.74 - 0.37)z^2 + (0.02 - (-0.50))z + (-0.36 - 0.14)$$

$$0.37z^2 + 0.53z - 0.50 = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad a = 0.37, \quad b = 0.53, \quad c = -0.50:$$

$$b^2 = 0.53^2 = 0.28 \quad 4ac = 4 \cdot 0.37 \cdot (-0.50) = -0.73$$

$$b^2 - 4ac = 0.28 - (-0.73) = 1.01 \quad \sqrt{1.01} = 1.00$$

$$z = \frac{-0.53 \pm 1.00}{2 \cdot 0.37} = \frac{-0.53 \pm 1.00}{0.74}$$

$$z_{b1} = \frac{-0.53 + 1.00}{0.74} = \frac{0.48}{0.74} = 0.65 \quad z_{b2} = \frac{-0.53 - 1.00}{0.74} = \frac{-1.53}{0.74} = -2.08$$

$$z_{b1} = 0.65 \quad (k \approx 0.20) \quad z_{b2} = -2.08 \quad (k \approx 15.03)$$

e) Cerc & trasare

$$z = z_0 + w, \quad |w|^2 = (z_0 - p_1)(z_0 - p_2)$$

$$r = |w| = \sqrt{(z_0 - p_1)(z_0 - p_2)}$$

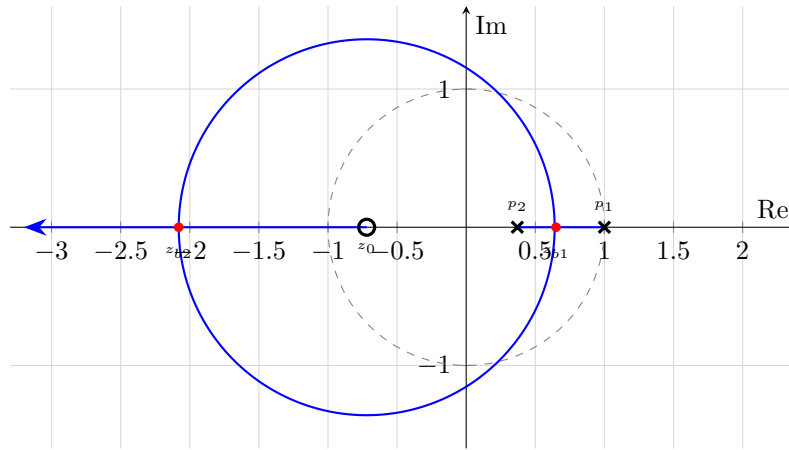
$$z_0 - p_1 = -0.72 - 1 = -1.72 \quad z_0 - p_2 = -0.72 - 0.37 = -1.09$$

$$r = \sqrt{(-1.72)(-1.09)} = \sqrt{1.86}$$

$$r = 1.36$$

$$w_{b1} = 0.65 - (-0.72) = 1.36 \quad w_{b2} = -2.08 - (-0.72) = -1.36$$

$$|w_{b1}| = |w_{b2}| = 1.36 = r$$



(cerc punctat = cerc unitate.)

Teorie (pe scurt)

1. Transformarea $s \rightarrow z$ (porțiunea 5-6)

$$z = e^{sT_e} : \quad \text{Re}(s) = 0 \rightarrow |z| = 1; \quad \text{Re}(s) < 0 \rightarrow |z| < 1 \quad \Rightarrow \text{aliasing}$$

2. Frecvența de eșantionare (aliasing)

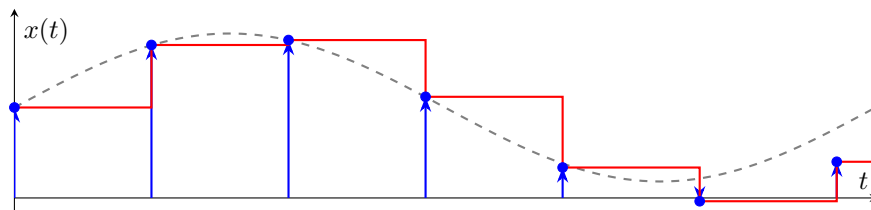
$$\omega_e > 2\omega_{max} \quad (\text{Nyquist-Shannon})$$

3. Etapele conversiei A/N

Eșantionare $\rightarrow$ Reținere (S&H) $\rightarrow$ Cuantizare $\rightarrow$ Codificare

4. Eșantionare + reconstrucție cu ERO

$$x^*(t) = \sum_{k=0}^{\infty} x(kT_e) \delta(t - kT_e) \quad \hat{x}(t) = x(kT_e), \quad kT_e \leq t < (k+1)T_e \quad G_{ERO}(s) = \frac{1 - e^{-sT_e}}{s}$$



(gri = $x(t)$; săgeți = $x^*(t)$; roșu = $\hat{x}(t)$ reconstruit cu ERO.)